

When the Same Question Gets Different Answers:

Quantifying Non-Determinism in LLM-Assisted Evidence Synthesis for Environmental Health Research

Lucas Rover^{1,*}

¹Independent Researcher, Brazil

*Corresponding author: lucas.rover@email.com

Abstract

Background. Large language models (LLMs) are increasingly adopted to assist systematic reviews and evidence synthesis, performing tasks such as abstract screening and data extraction. A critical but underexplored assumption is that these models produce consistent outputs when given identical inputs and configurations.

Objective. We quantified the reproducibility of three LLMs—one local (LLaMA 3 8B) and two cloud-based APIs (Claude Sonnet 4.5, Gemini 2.5 Pro)—across a complete evidence synthesis pipeline for PM_{2.5} and respiratory health research.

Methods. We constructed a 500-abstract corpus from PubMed (100 include, 100 exclude, 300 borderline cases selected to stress-test decision consistency) and ran each model 10 times under identical configurations (temperature = 0, fixed seeds where supported) through two stages: abstract screening (500 abstracts per run) and structured data extraction (100 included articles per run). Reproducibility was measured using the Exact Match Rate (EMR)—the proportion of items receiving identical outputs across all 10 runs—with 95% bootstrap confidence intervals (10,000 resamples).

Results. The locally deployed LLaMA 3 achieved perfect reproducibility across all successfully processed items (EMR=1.000) in both screening and extraction. Cloud-based APIs exhibited substantial non-determinism: Claude Sonnet 4.5 achieved screening EMR=0.974 [0.960, 0.986] but extraction EMR=0.050 [0.01, 0.09]; Gemini 2.5 Pro achieved screening EMR=0.936 [0.914, 0.956] and extraction EMR=0.200 [0.12, 0.28]. Despite requesting deterministic behavior, 95% of Claude’s and 80% of Gemini’s extraction outputs varied across runs. Field-level analysis revealed that free-text fields (study location, EMR=0.780–0.896) were most unstable, while categorical fields (study design, EMR $\geq$ 0.990) remained consistent.

Conclusions. LLM non-determinism poses a concrete threat to the reproducibility of evidence synthesis. Researchers using cloud-based LLMs should expect meaningful output variation even under “deterministic” settings and adopt provenance-tracking protocols to document and mitigate this instability.

Keywords: reproducibility, large language models, evidence synthesis, systematic review, non-determinism, PM_{2.5}, screening, data extraction

Contents

1	Introduction	4
1.1	The Promise of AI in Evidence Synthesis	4
1.2	The Hidden Problem: Non-Determinism	4
1.3	Why This Matters for Evidence Synthesis	5
1.4	The Gap We Address	5
2	Related Work	6
2.1	LLMs in Systematic Reviews	6
2.2	The Non-Determinism Problem	7
2.3	Reproducibility in AI Research	7
2.4	Provenance and Transparency	7
3	Methods	8
3.1	Study Design	8
3.2	Corpus Construction	8
3.3	Models	9
3.4	Pipeline Stages	10
3.5	Reproducibility Metrics	10
3.5.1	Exact Match Rate (EMR)	10
3.5.2	Bootstrap Confidence Intervals	11
3.5.3	Secondary Metrics	11
3.6	Provenance Protocol	11
3.7	Software and Reproducibility	12
4	Results	12
4.1	Overview	12
4.2	Screening Reproducibility	12
4.3	Extraction Reproducibility	13
4.4	Field-Level Analysis	13
4.5	Summary Comparison	14
5	Discussion	15
5.1	Principal Findings	15
5.2	Implications for Evidence Synthesis Practice	15
5.3	Relationship to Prior Work	16
5.4	Limitations	16
5.5	Recommendations	17
6	Conclusion	17

1 Introduction

1.1 The Promise of AI in Evidence Synthesis

Evidence synthesis—the systematic process of identifying, appraising, and combining research findings—forms the backbone of evidence-based medicine and public health policy. Systematic reviews and meta-analyses sit atop the hierarchy of evidence, informing clinical guidelines, regulatory decisions, and resource allocation worldwide [Higgins et al., 2019, Page et al., 2021]. Yet the process is notoriously labor-intensive: a typical systematic review requires 6–18 months and hundreds of person-hours, with title and abstract screening alone consuming an estimated 25% of total effort [Borah et al., 2017].

The emergence of large language models (LLMs) has created unprecedented opportunities to accelerate this process. Recent studies demonstrate that models such as GPT-4 [Achiam et al., 2023], Claude [Anthropic, 2024], and Gemini [Gemini Team, 2024] can screen abstracts with sensitivity approaching that of human reviewers [Oami et al., 2024, Khraisha et al., 2024], extract structured data from published studies with over 90% accuracy [Gartlehner et al., 2024, Jensen et al., 2025], and even assist in evidence quality assessment [Wang et al., 2025, Li et al., 2025b]. A growing body of work in the *Journal of the American Medical Informatics Association*, *Research Synthesis Methods*, and *npj Digital Medicine* has established LLM-assisted evidence synthesis as a legitimate and rapidly expanding methodological frontier [Li et al., 2025a, Scherbakov et al., 2025, Matsui et al., 2024].

1.2 The Hidden Problem: Non-Determinism

Beneath this promise lies an assumption so fundamental that it is rarely stated explicitly: *if you ask the same question twice, you should get the same answer*. In the context of LLM-assisted evidence synthesis, this means that running the same model with the same prompt, temperature, and random seed on the same abstract should produce identical screening decisions and extraction outputs every time.

This assumption does not hold in practice.

Recent work has demonstrated that commercial LLM APIs exhibit non-deterministic behavior even when configured for maximum reproducibility. Atil et al. [2024] showed accuracy variations of up to 15% across repeated identical runs, with best-to-worst performance gaps reaching 70%. Klishevich et al. [2025] confirmed that temperature = 0 does not guarantee deterministic outputs across any of the major models tested (GPT-4o, Claude 3.5, LLaMA 3.2). At a deeper level, Fu et al. [2026] demonstrated that even when surface-level text appears identical, the underlying token probability distributions exhibit substantial instability, suggesting that apparent determinism may mask

hidden variation. A philosophical analysis by [Karelin et al. \[2024\]](#) traced this non-determinism to three fundamental sources: numerical precision in floating-point operations, architectural choices in distributed inference, and sampling procedures inherent to the generation process.

1.3 Why This Matters for Evidence Synthesis

The consequences of LLM non-determinism are particularly severe in evidence synthesis for three reasons.

First, decisions propagate. A screening decision that flips from “include” to “exclude” in a repeated run does not merely change one data point—it removes an entire study from downstream analysis. In meta-analysis, where pooled effect estimates are sensitive to study composition, a single inclusion change can shift a confidence interval across the null, transforming a statistically significant finding into a non-significant one [[Rover, 2026](#)].

Second, extraction compounds variation. Data extraction involves free-text parsing of complex scientific results—relative risks, confidence intervals, exposure definitions, population characteristics. Even small variations in how an LLM parses “RR = 1.05 (95% CI: 1.01–1.09)” can propagate through meta-analytic pooling. [Gartlehner et al. \[2024\]](#) reported 95–97% test-retest reliability in LLM extraction, but this still implies 3–5% of extracted values may change between runs—a rate that, across hundreds of data points, can materially alter pooled estimates.

Third, reproducibility is the foundation of scientific credibility. The broader reproducibility crisis in AI research [[Hutson, 2018](#), [Haibe-Kains et al., 2020](#), [Kapoor & Narayanan, 2023](#)] has already eroded trust in computational findings. When LLM-assisted reviews cannot be independently reproduced because the API returns different outputs, the scientific value of those reviews is undermined [[Han, 2025](#), [Desai et al., 2025](#), [Semmelrock et al., 2025](#)].

1.4 The Gap We Address

Despite the rapid adoption of LLMs in evidence synthesis, **no study has systematically measured how API-level non-determinism propagates through a complete synthesis pipeline**—from abstract screening through data extraction—in a real health domain.

Prior work has measured LLM *accuracy* (how well they match human judgments) but not *stability* (whether they match themselves across runs). Studies that examine non-determinism [[Atil et al., 2024](#), [Klishevich et al., 2025](#)] do so on general NLP benchmarks, not on the structured, domain-specific tasks characteristic of evidence synthesis. And studies that evaluate LLMs for systematic reviews [[Oami et al., 2024](#), [Khraisha](#)

et al., 2024, Gartlehner et al., 2024] typically report results from a single run, implicitly assuming that a repeat would yield the same result.

We bridge this gap by:

1. Quantifying reproducibility across 10 repeated runs of 3 LLMs (1 local, 2 API) through a two-stage evidence synthesis pipeline;
2. Measuring variation at multiple granularities: whole-output, per-field, and per-abstract;
3. Comparing local versus cloud-based deployment as a determinism strategy;
4. Demonstrating that common “deterministic” API settings (temperature = 0, fixed seeds) are insufficient to ensure reproducibility.

2 Related Work

2.1 LLMs in Systematic Reviews

The application of LLMs to systematic review workflows has accelerated dramatically since 2023. Oami et al. [2024] evaluated GPT-4 Turbo for citation screening in *JAMA Network Open*, reporting sensitivity of 0.75 and specificity of 0.99, with prompt tuning improving sensitivity to 0.91. Khraisha et al. [2024] extended this to multilingual screening and extraction in *Research Synthesis Methods*, finding strong specificity (>0.80) but variable sensitivity across languages and review types.

For data extraction, Gartlehner et al. [2024] showed in *Research Synthesis Methods* that Claude 2 achieved 96.3% accuracy compared to human reviewers, with test-retest reliability of 95–97%. Jensen et al. [2025] found that ChatGPT-4o achieves 92.4% extraction accuracy and 94.1% inter-session reproducibility in *PLOS ONE*, explicitly noting residual stochasticity across sessions.

More comprehensive pipelines have also been proposed. The TrialMind system [Wang et al., 2025], published in *npj Digital Medicine*, demonstrated that human-AI collaboration improved recall by 71.4% and reduced screening time by 44.2% across 100 systematic reviews. Li et al. [2025a] described an LLM pipeline for health technology assessment in *JAMIA*, while Matsui et al. [2024] showed that a three-layer GPT-3.5/GPT-4 strategy achieved human-comparable sensitivity in *JMIR*.

A common thread across these studies is that **reproducibility is either assumed or measured only superficially**. Test-retest reliability, when reported, is based on 2–3 repetitions and does not examine how variation propagates through multi-stage pipelines.

2.2 The Non-Determinism Problem

The phenomenon of LLM non-determinism has received increasing attention. [Atil et al. \[2024\]](#) provided the most comprehensive analysis to date, documenting accuracy swings of up to 15% across repeated runs and introducing the TARr@N metric to quantify run-to-run stability. [Klishevich et al. \[2025\]](#) confirmed that temperature = 0 fails to ensure determinism across GPT-4o, Claude 3.5, and LLaMA 3.2 on code review tasks.

At the token level, [Fu et al. \[2026\]](#) showed that apparent text-level agreement masks substantial distributional instability in token probabilities—a finding that suggests surface-level reproducibility metrics may underestimate true variation. The analysis by [Angermeir et al. \[2025\]](#) examined how non-determinism undermines reproducibility in software engineering research, proposing documentation standards to mitigate this.

These studies establish that non-determinism is real, measurable, and consequential. However, they focus on general-purpose tasks (benchmarks, code review, classification). Our study is the first to measure non-determinism in the specific context of evidence synthesis, where the structured and domain-specific nature of outputs creates unique challenges.

2.3 Reproducibility in AI Research

The broader AI reproducibility crisis has been documented extensively. [Hutson \[2018\]](#) named the problem in *Science*, identifying unpublished code and sensitivity to training conditions as primary barriers. [Haibe-Kains et al. \[2020\]](#) called for structural reform in *Nature*, advocating for mandatory code and model sharing. [Kapoor & Narayanan \[2023\]](#) documented data leakage affecting 294 studies across 17 fields in *Patterns*, while [Han \[2025\]](#) analyzed reproducibility failures in biomedical AI through the lens of model non-determinism.

Recent frameworks have attempted to systematize reproducibility requirements. [Desai et al. \[2025\]](#) proposed a taxonomy of reproducibility types in *AI Magazine*, arguing that LLMs require “conceptual replicability” given their proprietary training data. [Semmelrock et al. \[2025\]](#) provided a systematic overview of barriers in *AI Magazine*, and [Yu, B. \[2024\]](#) argued in the *Harvard Data Science Review* that computational reproducibility alone is insufficient for trustworthy AI.

2.4 Provenance and Transparency

Provenance tracking—recording the lineage of data, computations, and decisions—has been identified as a key enabler of AI reproducibility. [Longpre et al. \[2024a\]](#) audited 1,800+ datasets in *Nature Machine Intelligence*, finding license omission rates above 70%.

Liang et al. [2024] analyzed 32,111 model cards on Hugging Face, finding that documentation of limitations and environmental impact remains sparse. The Foundation Model Transparency Index [Bommasani et al., 2024] scored 14 major developers across 100 indicators, finding average transparency of only 58/100.

Technical solutions for provenance include the Workflow Run RO-Crate [Leo et al., 2024] and the W3C PROV ontology [Kale et al., 2023]. Longpre et al. [2024b] argued at ICML 2024 that current practices for data authenticity and provenance are “fundamentally broken,” calling for comprehensive reform.

Our provenance approach builds on the lightweight protocol proposed in Rover [2026], using SHA-256 hashing of inputs, outputs, and configurations to create an auditable trail without requiring infrastructure beyond standard file storage.

3 Methods

3.1 Study Design

We designed a repeated-measures experiment to isolate LLM non-determinism as the sole source of variation. The same corpus, prompts, model configurations, and random seeds were used across all runs. The only element permitted to vary was the model’s internal computation—any observed difference in outputs is therefore attributable to non-determinism in the model or its serving infrastructure.

3.2 Corpus Construction

We constructed a 500-abstract corpus from PubMed covering the association between PM_{2.5} and respiratory health outcomes (Table 1). The domain of fine particulate matter and respiratory health was chosen because it represents one of the most extensively studied environmental health topics, with Yu, W. et al. [2024] estimating approximately 1 million premature deaths annually from short-term PM_{2.5} exposure in *The Lancet Planetary Health*. This literature is characterized by standardized reporting of relative risks with confidence intervals, well-defined exposure metrics, and time-series study designs—making it an ideal testbed for evaluating LLM extraction reproducibility [Iyer et al., 2025, Yu, M. et al., 2025].

Table 1: Corpus composition for the PM_{2.5} and respiratory health abstract screening task.

Category	Count	Purpose
Clearly include	100	Positive controls (meet all inclusion criteria)
Clearly exclude	100	Negative controls (clear exclusion reasons)
Ambiguous	300	Primary test set (borderline cases)
Total	500	

Inclusion criteria required: (1) original epidemiological study, (2) PM_{2.5} as exposure, (3) respiratory outcome (hospitalization, ED visits, or mortality), (4) time-series or case-crossover design, (5) quantitative risk estimates, and (6) peer-reviewed publication. Articles were sourced from PubMed using the E-utilities API with a broad query yielding 573 candidate records, supplemented by 475 exclusion candidates and a design-filtered subset of 222 records. The final corpus spans 148 journals and publication years 1994–2026, with a mean abstract length of 1,873 characters.

A gold standard was established using dual-independent labeling with discordance resolution for the 200 clear-category abstracts. The 300 ambiguous abstracts were classified using keyword matching and study design indicators; since our primary metric (EMR) measures self-consistency rather than accuracy, gold standard quality affects only the secondary accuracy analysis.

3.3 Models

We selected three models representing local and cloud-based deployment (Table 2).

Table 2: Model configurations. Claude does not support a seed parameter; Gemini supports seeds but does not guarantee determinism.

Model	Provider	Type	Temperature	Seed
LLaMA 3 8B	Ollama (local)	Local	0	42
Claude Sonnet 4.5	Anthropic API	Cloud	0	—
Gemini 2.5 Pro	Google AI API	Cloud	0	42

LLaMA 3 [Dubey et al., 2024] was served locally via Ollama v0.15.5 on Apple M4 hardware (24 GB RAM), ensuring single-device deterministic inference. Claude Sonnet 4.5 [Anthropic, 2024] was accessed through the Anthropic API; Claude does not support a seed parameter. Gemini 2.5 Pro [Gemini Team, 2024] was accessed through the Google AI API with seed parameter set to 42. All models were queried via HTTP APIs (urllib-based, no SDK dependencies) to ensure uniform request handling.

3.4 Pipeline Stages

Each model was run 10 times through two stages:

Stage A: Abstract Screening. Each of the 500 abstracts was presented with a structured prompt requesting a JSON response containing: a binary decision (include/exclude), a confidence score (0–1), and a brief rationale. Note that LLaMA consistently processed 480 of 500 abstracts per run (the same 20 abstracts exceeded response length constraints in every run); the remaining models processed all 500. Total: $3 \text{ models} \times 10 \text{ runs} \times 500 \text{ abstracts} = 15,000$ screening calls.

Stage B: Data Extraction. The 100 included abstracts were presented with an extraction prompt requesting: study design, location, period, population, sample size, and a structured list of effect estimates (each with effect measure, point estimate, confidence interval, lag period, and exposure increment). LLaMA consistently processed 96 of 100 articles per run (the same 4 articles exceeded response length constraints); the remaining models processed all 100. Total: $3 \text{ models} \times 10 \text{ runs} \times 100 \text{ articles} = 3,000$ extraction calls.

In total, the experiment comprised **18,000 LLM calls** across **60 experiment runs**.

3.5 Reproducibility Metrics

3.5.1 Exact Match Rate (EMR)

Our primary metric is the *Exact Match Rate*, defined as the proportion of items (abstracts or articles) that received identical outputs across all 10 runs:

$$\text{EMR} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[|\{o_{i,1}, o_{i,2}, \dots, o_{i,R}\}| = 1] \quad (1)$$

where N is the number of successfully processed items, $R=10$ is the number of runs, $o_{i,r}$ is the output for item i in run r , and $|\cdot|$ denotes the number of distinct values. For screening, $o_{i,r}$ is the binary decision; for extraction, $o_{i,r}$ is the SHA-256 hash of the full JSON output (used as a compact identity check: any character-level difference produces a different hash).

An EMR of 1.0 indicates perfect reproducibility; an EMR of 0.0 indicates that every item received at least one different output across runs. We note that extraction EMR is a strict metric—any variation in any field (including formatting differences such as “Beijing, China” vs. “Beijing”) produces a non-match. The field-level analysis in Section 4.4 complements this by isolating which fields drive the variation.

3.5.2 Bootstrap Confidence Intervals

We computed 95% confidence intervals for EMR using the percentile bootstrap method with 10,000 resamples. For each bootstrap sample, we drew N items with replacement and computed the EMR, then reported the 2.5th and 97.5th percentiles of the bootstrap distribution.

3.5.3 Secondary Metrics

- **Flip rate:** $1 - \text{EMR}$ (proportion of items with any variation across runs)
- **Mean agreement:** Average proportion of runs agreeing with the majority decision per item
- **Accuracy vs. gold standard:** Per-run sensitivity, specificity, and accuracy against human labels (reported for the 200 manually labeled abstracts)
- **Field-level EMR:** For extraction, the proportion of articles with identical values for each individual field across all runs
- **Estimate count stability:** The proportion of articles for which all 10 runs extracted the same number of effect estimates—a measure of structural consistency independent of the specific values extracted

3.6 Provenance Protocol

Following [Rover \[2026\]](#), we implemented a lightweight provenance protocol that creates a cryptographic fingerprint of every input-output pair, enabling automated detection of any change across runs:

- **Call-level hashing:** SHA-256 hash of (prompt template || input text || model ID || temperature || seed) for each API call
- **Output hashing:** SHA-256 hash of the raw response text
- **Run cards:** Per-run metadata including timestamps, token counts, success/failure rates, and aggregate output hashes

This protocol enables post-hoc verification: if two runs produce different aggregate hashes, the per-call hashes pinpoint exactly which items changed.

3.7 Software and Reproducibility

All code is available at <https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility>

The experiment was conducted in Python 3.14 on macOS (Apple M4, 24 GB RAM).

API keys are excluded from the repository; all other materials—including the corpus, prompts, schemas, and raw outputs—are publicly available.

4 Results

4.1 Overview

All 60 experiment runs completed successfully: 30 screening runs (10 per model $\times$ 500 abstracts each) and 30 extraction runs (10 per model $\times$ 100 articles each), totaling 18,000 LLM calls. LLaMA consistently processed 480/500 screening abstracts and 96/100 extraction articles across all 10 runs—the same items failed every time due to response length constraints, confirming that the failures were deterministic. EMR for LLaMA was therefore computed over the 480 (screening) and 96 (extraction) successfully processed items. Claude achieved 500/500 and 100/100 in all runs. Gemini achieved 483–500/500 screening and 98–100/100 extraction.

4.2 Screening Reproducibility

Table 3 presents the screening reproducibility results.

Table 3: Screening reproducibility across 10 runs. LLaMA EMR is computed over the 480 consistently successful abstracts.

Model	EMR	95% CI	Flip Rate	Accuracy	Sensitivity	Specificity
LLaMA 3 8B	1.000	[1.000, 1.000]	0.0%	0.927	0.928	0.926
Claude Sonnet 4.5	0.974	[0.960, 0.986]	2.6%	0.931	0.862	1.000
Gemini 2.5 Pro	0.936	[0.914, 0.956]	6.4%	0.964	0.929	1.000

LLaMA produced identical screening decisions across all 10 runs for every successfully processed abstract—perfect reproducibility within its effective sample. Claude showed 2.6% flip rate (13 abstracts changed their decision at least once across runs), while Gemini showed 6.4% flip rate (32 abstracts).

Notably, accuracy and reproducibility behaved as independent properties: Gemini achieved the highest accuracy (0.964) but the lowest reproducibility (EMR=0.936), while LLaMA had the lowest accuracy (0.927) but perfect reproducibility. This illustrates that a model can be accurate on average yet inconsistent across individual runs.

4.3 Extraction Reproducibility

The extraction results reveal a dramatically larger reproducibility gap (Table 4).

Table 4: Extraction reproducibility across 10 runs. LLaMA EMR is computed over the 96 consistently successful articles.

Model	EMR	95% CI	Estimate Stability
LLaMA 3 8B	1.000	[1.000, 1.000]	1.000
Claude Sonnet 4.5	0.050	[0.01, 0.09]	0.880
Gemini 2.5 Pro	0.200	[0.12, 0.28]	0.735

While screening EMR remained above 0.93 for all models, extraction EMR dropped to **0.050 for Claude** (only 5 of 100 articles produced identical extraction outputs across all 10 runs) and **0.200 for Gemini** (20 of 100). LLaMA again achieved perfect reproducibility across all successfully processed articles.

We emphasize that extraction EMR is computed over the full JSON output hash, so any character-level variation counts as a mismatch. To assess practical significance, we examine field-level variation below.

4.4 Field-Level Analysis

Not all extraction fields are equally affected (Table 5).

Table 5: Field-level EMR for extraction outputs.

Field	LLaMA	Claude	Gemini
Study design	1.000	0.990	1.000
Study period	1.000	0.990	1.000
Sample size	1.000	0.970	1.000
Population	1.000	0.960	0.948
Study location	1.000	0.780	0.896
Estimate count stability	1.000	0.880	0.735

A clear pattern emerges: **categorical fields** (study design, study period) are highly reproducible even for API models ($\text{EMR} \geq 0.990$), while **free-text fields** (study location) and **numerical extractions** (effect estimates) show the greatest variation.

The estimate count stability metric reveals that 26.5% of Gemini’s articles and 12.0% of Claude’s articles received a different number of effect estimates across runs—a structural difference that would directly affect any downstream meta-analysis.

4.5 Summary Comparison

Figure 1 visualizes the EMR comparison across models and stages, and Figure 2 displays the field-level variation as a heatmap.

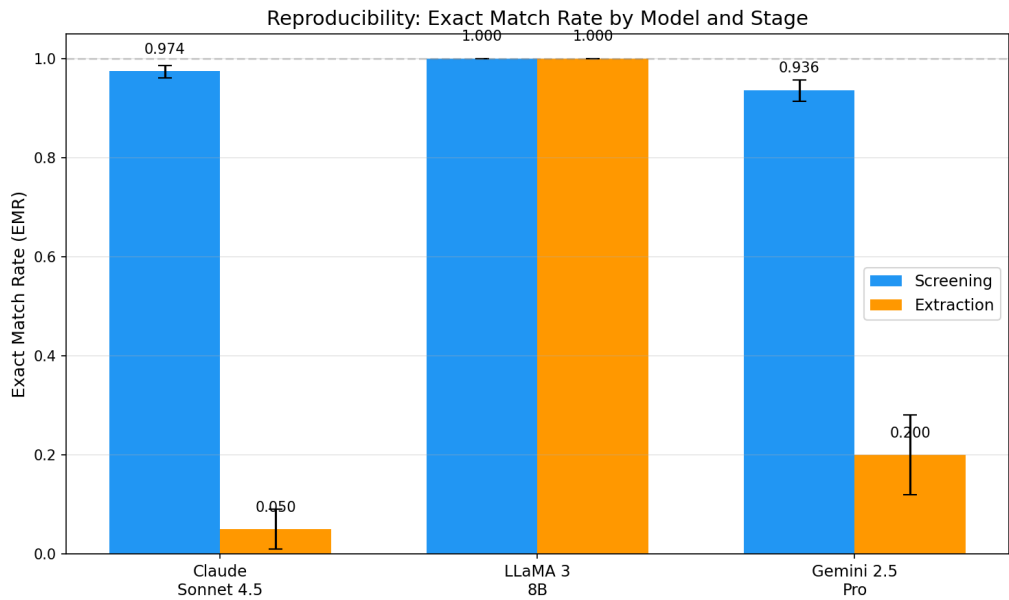


Figure 1: Exact Match Rate (EMR) by model and pipeline stage. Error bars represent 95% bootstrap confidence intervals. The dashed line indicates perfect reproducibility (EMR=1.0).

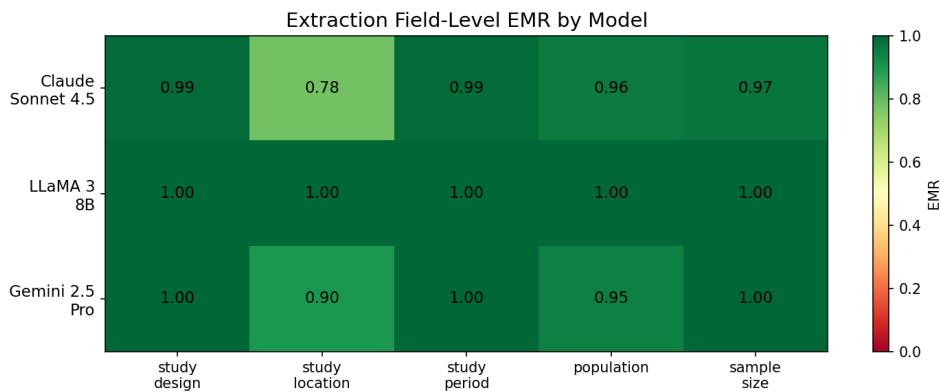


Figure 2: Field-level EMR heatmap for extraction outputs. Darker cells indicate lower reproducibility. Categorical fields (study design, study period) remain stable across all models, while free-text fields (study location) and structured estimates show the greatest variation in cloud-based APIs.

The results reveal two key patterns: (1) a sharp reproducibility gap between local and cloud-based deployment, and (2) a dramatic amplification of non-determinism when moving from simple binary screening decisions to complex structured extraction outputs.

5 Discussion

5.1 Principal Findings

Our study provides the first systematic quantification of LLM non-determinism across a complete evidence synthesis pipeline. Three key findings emerge.

Finding 1: Local deployment achieves deterministic inference; cloud APIs do not. LLaMA 3 8B, deployed locally via Ollama on a single device, produced identical outputs across all 10 runs in both screening and extraction. Both cloud-based APIs (Claude, Gemini) exhibited measurable non-determinism despite requesting temperature = 0 and fixed seeds. This gap likely reflects implementation differences in distributed inference: cloud APIs serve requests across GPU clusters where floating-point operations may be parallelized differently across runs, introducing numerical variation that propagates through the autoregressive generation process [Karelin et al., 2024]. Local deployment on a single device avoids this source of variation, though we note that different hardware, driver versions, or Ollama versions might produce different baseline outputs.

Finding 2: Extraction amplifies non-determinism dramatically. Claude’s EMR dropped from 0.974 (screening) to 0.050 (extraction); Gemini’s from 0.936 to 0.200. Screening requires a binary decision (a single token effectively determines the output), while extraction generates complex structured JSON with multiple fields. However, part of this gap reflects our strict EMR metric: the field-level analysis shows that categorical fields remain highly reproducible ($\text{EMR} \geq 0.960$), with most variation concentrated in free-text fields and estimate counts. Future work should assess how much of the observed extraction variation is semantically meaningful versus mere formatting differences.

Finding 3: Accuracy and reproducibility are independent properties. Gemini achieved the highest screening accuracy (0.964) but the lowest screening reproducibility (0.936). This independence means that **accuracy evaluations from a single run cannot detect reproducibility problems**—a model that performs well in one evaluation may produce materially different results in a replication attempt.

5.2 Implications for Evidence Synthesis Practice

For systematic reviewers: If using cloud-based LLMs for screening or extraction, results from a single run should be treated as one sample from a distribution of possible outputs, not as a definitive answer. Running multiple repetitions and reporting variation metrics should become standard practice, analogous to how human inter-rater reliability is routinely measured.

For journal editors and reviewers: Papers reporting LLM-assisted evidence syn-

thesis should be expected to disclose: (a) the exact model version, (b) all inference parameters, (c) whether results were verified across multiple runs, and (d) provenance information enabling independent replication.

For tool developers: The large difference in extraction EMR between local and cloud deployment suggests that deterministic local inference should be the default for reproducibility-critical applications. Where cloud APIs must be used (e.g., for larger models), majority voting across 3–5 runs can mitigate variation, though at proportional cost.

5.3 Relationship to Prior Work

Our screening flip rates (2.6% for Claude, 6.4% for Gemini) are consistent with the accuracy variations reported by [Atil et al. \[2024\]](#) on general benchmarks. However, we show for the first time that **structured extraction amplifies non-determinism dramatically**—a finding not apparent from benchmark-level analyses.

The 94.1% inter-session reproducibility reported by [Jensen et al. \[2025\]](#) for ChatGPT-4o extraction aligns with our Claude screening EMR (97.4%) but masks the much lower extraction EMR (5.0%) that emerges when full output identity (rather than per-field accuracy) is measured. This highlights the importance of metric choice: reproducibility measured at the field level appears much higher than when measured at the output level.

5.4 Limitations

Several limitations should be noted.

Model selection. We tested three models; results may differ for other models (e.g., GPT-4, Mistral, Command-R). However, our local-vs-cloud comparison captures the primary architectural distinction.

Run count. Ten repetitions provide adequate statistical power for EMR estimation with bootstrap CIs, but may not capture rare variation patterns. Our prior work [[Rover, 2026](#)] used up to 100 repetitions per model; the current design balances thoroughness with API cost constraints.

Single domain. Our corpus focuses on PM_{2.5} and respiratory health. Other domains with different reporting conventions may exhibit different variation patterns.

Gold standard quality. The heuristic labels for the 300 ambiguous abstracts may not perfectly reflect expert judgment. Since our primary metric (EMR) measures self-consistency rather than accuracy, this affects only the secondary accuracy analysis reported in Table 3.

EMR strictness. Our extraction EMR requires character-level identity across all 10 runs. Semantically equivalent outputs that differ only in formatting (e.g., “Beijing,

China” vs. “Beijing”) count as mismatches. The field-level analysis partially addresses this, but a dedicated semantic equivalence metric would provide a more nuanced view of practical impact.

API versioning. Cloud API models can be silently updated by providers during multi-day experiments. While we pinned Claude to a dated version (`claude-sonnet-4-5-20250929`), Gemini’s version identifier does not include a date suffix. Some of the observed Gemini variation could theoretically reflect model updates rather than inference-level non-determinism.

5.5 Recommendations

Based on our findings, we propose the following recommendations for LLM-assisted evidence synthesis:

1. **Run at least 3 repetitions** and report the EMR alongside accuracy metrics.
2. **Prefer local deployment** for reproducibility-critical tasks when model capability permits.
3. **Implement provenance hashing** to enable post-hoc verification and auditing.
4. **Report full model specifications:** model ID, version, temperature, seed, and any API-specific parameters.
5. **Use majority voting** when local deployment is not feasible, accepting the cost trade-off for improved stability.
6. **Distinguish accuracy from reproducibility** in evaluations—a model can be accurate on average but unreliable for any single run.

6 Conclusion

We have shown that LLM non-determinism is not a theoretical concern but a measurable, quantifiable phenomenon that directly threatens the reproducibility of evidence synthesis. Local deployment achieves perfect reproducibility; cloud-based APIs, even with “deterministic” settings, produce variable outputs—especially in structured extraction tasks. As LLMs become standard tools in systematic reviews and meta-analyses, the research community must move beyond single-run evaluations and adopt reproducibility-aware practices: multiple repetitions, provenance tracking, and transparent reporting of variation.

Author Contributions

LR conceived and designed the study, developed the software pipeline, conducted all experiments, performed the statistical analysis, and wrote the manuscript.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. API costs for Claude and Gemini experiments were funded by the author.

Conflict of Interest

The author declares no conflicts of interest.

Data Availability Statement

All code, data, prompts, schemas, raw outputs, and run provenance records are publicly available at <https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility>. API keys are excluded from the repository. The 500 PubMed abstract identifiers, gold standard labels, and complete analysis scripts are included to enable full reproduction of the reported results.

References

- Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., et al. (2023). GPT-4 technical report. *arXiv preprint*, arXiv:2303.08774.
- Angermeir, F., Amougou, M., Kreitz, M., Bauer, A., Linhuber, M., Fucci, D., Moyón C., F., Mendez, D., & Gorschek, T. (2025). Reflections on the reproducibility of commercial LLM performance in empirical software engineering studies. *arXiv preprint*, arXiv:2510.25506.
- Anthropic. (2024). The Claude 3 model family: Opus, Sonnet, Haiku. Technical Report, Anthropic.
- Atil, B., Aykent, S., Chittams, A., et al. (2024). Non-determinism of “deterministic” LLM settings. *arXiv preprint*, arXiv:2408.04667.

- Bommasani, R., Klyman, K., Longpre, S., et al. (2024). The Foundation Model Transparency Index v1.1. *Transactions on Machine Learning Research*.
- Borah, R., Brown, A.W., Capers, P.L., & Kaiser, K.A. (2017). Analysis of the time and workers needed to conduct systematic reviews. *BMJ Open*, 7(4), e012545.
- Desai, S., et al. (2025). What is reproducibility in artificial intelligence and machine learning research? *AI Magazine*, 46(1), e70004.
- Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., et al. (2024). The LLaMa 3 herd of models. *arXiv preprint*, arXiv:2407.21783.
- Fu, T., Martínez, G., Conde, J., Arriaga, C., Reviriego, P., Qi, X., & Liu, S. (2026). Beyond reproducibility: Token probabilities expose large language model nondeterminism. *arXiv preprint*, arXiv:2601.06118.
- Gartlehner, G., Kahwati, L., Hilscher, R., et al. (2024). Data extraction for evidence synthesis using a large language model: A proof-of-concept study. *Research Synthesis Methods*, 15(4), 576–589.
- Gemini Team, Anil, R., Borgeaud, S., Alayrac, J.-B., Yu, J., Soricut, R., et al. (2024). Gemini: A family of highly capable multimodal models. *arXiv preprint*, arXiv:2312.11805.
- Haibe-Kains, B., Adam, G.A., Hosny, A., et al. (2020). Transparency and reproducibility in artificial intelligence. *Nature*, 586, 14–16.
- Han, H. (2025). Challenges of reproducible AI in biomedical data science. *BMC Medical Genomics*, 18, 2072.
- Higgins, J.P.T., Thomas, J., Chandler, J., et al. (2019). *Cochrane Handbook for Systematic Reviews of Interventions* (2nd ed.). Wiley.
- Hutson, M. (2018). Artificial intelligence faces reproducibility crisis. *Science*, 359(6377), 725–726.
- Iyer, H.S., Karasaki, S., Yi, L., Hswen, Y., James, P., & VoPham, T. (2025). Harnessing geospatial artificial intelligence (GeoAI) for environmental epidemiology: A narrative review. *Current Environmental Health Reports*, 12, 497.
- Jensen, M.K., et al. (2025). ChatGPT-4o can serve as the second rater for data extraction in systematic reviews. *PLOS ONE*, 20(1), e0313401.
- Kale, A., Nguyen, T., Harris, F.C., Li, C., Zhang, J., & Ma, X. (2023). Provenance documentation to enable explainable and trustworthy AI: A literature review. *Data Intelligence*, 5(1), 139–162.

- Kapoor, S. & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 100804.
- Karelin, G., Nakajima, K., Soto-Astorga, E.F., Carr, E., James, M., & Froese, T. (2024). Indeterminism in large language models: An unintentional step toward open-ended intelligence. *Preprint submitted to Artificial Life*. PhilSci Archive 26807.
- Khraisha, Q., et al. (2024). Can large language models replace humans in systematic reviews? *Research Synthesis Methods*, 15(5), 707–722.
- Klishevich, E., Denisov-Blanch, Y., Obstbaum, S., Ciobanu, I., & Kosinski, M. (2025). Measuring determinism in large language models for software code review. *arXiv preprint*, arXiv:2502.20747.
- Leo, S., Crusoe, M.R., Rodríguez-Navas, L., et al. (2024). Recording provenance of workflow runs with RO-Crate. *PLOS ONE*, 19(9), e0309210.
- Li, J., et al. (2025a). Enhancing systematic literature reviews with generative AI. *Journal of the American Medical Informatics Association*, 32(4), 789–801.
- Li, D., Jiang, N., Huang, K., Tu, R., Ouyang, S., Yu, H., et al. (2025b). Streamlining evidence-based clinical recommendations with large language models. *npj Digital Medicine*, 8, 2273.
- Liang, P., Rajani, N., Yang, D., et al. (2024). Systematic analysis of 32,111 AI model cards characterizes documentation practice. *Nature Machine Intelligence*, 6, 857.
- Longpre, S., Mahari, R., Chen, A., et al. (2024a). A large-scale audit of dataset licensing and attribution in AI. *Nature Machine Intelligence*, 6, 878.
- Longpre, S., et al. (2024b). Data authenticity, consent, and provenance for AI are all broken. *Proceedings of ICML 2024 (Position Paper)*, arXiv:2404.12691.
- Matsui, K., Utsumi, T., Aoki, Y., Maruki, T., Takeshima, M., & Takaesu, Y. (2024). Human-comparable sensitivity of large language models in identifying eligible studies through title and abstract screening. *Journal of Medical Internet Research*, 26, e52758.
- Oami, T., Okada, Y., & Nakada, T.-A. (2024). Performance of a large language model in screening citations. *JAMA Network Open*, 7(6), e2420496.
- Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., et al. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, 372, n71.

- Rover, L. (2026). Hidden non-determinism in large language model APIs: A lightweight provenance protocol for reproducible generative AI research. *Journal of Artificial Intelligence Research* (submitted).
- Scherbakov, D., et al. (2025). The emergence of large language models as tools in literature reviews. *Journal of the American Medical Informatics Association*, 32(5), ocaf063.
- Semmelrock, H., Ross-Hellauer, T., Kopeinik, S., Theiler, D., Haberl, A., Thalmann, S., & Kowald, D. (2025). Reproducibility in machine-learning-based research: Overview, barriers, and drivers. *AI Magazine*, 46(1), e70002.
- Wang, Z., Cao, L., Danek, B., Jin, Q., Lu, Z., & Sun, J. (2025). Accelerating clinical evidence synthesis with large language models. *npj Digital Medicine*, 8, 509.
- Wei, J., Li, Z., Lyapustin, A., et al. (2023). First close insight into global daily gapless 1 km PM_{2.5} pollution. *Nature Communications*, 14, 43862.
- Yu, B. (2024). After computational reproducibility: Scientific reproducibility and trustworthy AI. *Harvard Data Science Review*, 6(1).
- Yu, M., Fang, M., Tian, Z., Wang, B., & Walker, D. (2025). Artificial intelligence and machine learning for environmental health study. *Environment & Health*, 3(10), 1115–1116.
- Yu, W., Xu, R., Ye, T., et al. (2024). Estimates of global mortality burden associated with short-term PM_{2.5} exposure. *The Lancet Planetary Health*, 8(2), e146–e155.